III B.TECH – I SEMESTER

Full Stack Development – Ii

Skill Enhancement Course | JNTUA R23 · B.Tech CSE

	FULL STACK DEVELOPMENT – II (Skill Enhancement Course)	L	T	P	C
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Course Objectives: The main objectives of the course are to

- Make use of Modern- day JavaScript with ES6 standards for designing Dynamic web pages
- Building robust & responsive User Interfaces using popular JavaScript library '**React.js**'. Building robust backend APIs using '**Express. js**'
- Establishing the connection between frontend (React) User interfaces and backend APIs (Express) with Data Bases(My SQL)

Course Out comes:

- CO1: Building fast and interactive UIs
- CO2: Applying Declarative approach for developing web apps
- CO3: Understanding ES6 features to embrace modern JavaScript
- CO4: Building reliable APIs with Express. js

Experiments covering the Topics:

- Introduction to DOM (Document Object Model), Ecma Script (ES6) standards and features like Arrow functions, Spread operator, Rest operator, Type coercion, Type hoisting, String literals, Array and Object Destructuring.
- Basics of React. js like React Components, JSX, Conditional rendering

Differences between Real DOM and Virtual DOM.

- Important React.js concepts like React hooks, Props, React forms, Fetch API, Iterative rendering using JavaScript map() function.
- JavaScript runtime environment node. js and its uses, Express. js and Routing, Micro-Services architecture and MVC architecture, database connectivity using (My SQL)
- Introduction to My SQL, setting up MySQL and configuring, Databases, My SQL queries, subqueries, creating My SQL driver for database connectivity to Express. js server.

Sample Experiments:

- **Introduction to Modern JavaScript and DOM**
- Write a JavaScript program to link JavaScript file with the HTML page
- Write a JavaScript program to select the elements in HTML page using selectors
- Write a JavaScript program to implement the event listeners
- Write a JavaScript program to handle the click events for the HTML button elements
- Write a JavaScript program to With three types of functions
- Function declaration
- Function definition
- Arrow functions
- **Basics of React. js**
- Write a React program to implement a counter button using react class components
- Write a React program to implement a counter button using react functional components
- Write a React program to handle the button click events in functional component
- Write a React program to conditionally render a component in the browser
- Write a React program to display text using String literals
- **Important concepts of React. js**
- Write a React program to implement a counter button using React use State hook
- Write a React program to fetch the data from an API using React use Effect hook
- Write a React program with two react components sharing data using Props.
- Write a React program to implement the forms in react

- Write a React program to implement the iterative rendering using map() function.
- **Introduction to Node. js and Express. js**
- Write a program to implement the ‘hello world’ message in the route through the browser using Express
- Write a program to develop a small website with multiple routes using Express. js
- Write a program to print the ‘hello world’ in the browser console using Express. js
- Write a program to implement the CRUD operations using Express. js
- Write a program to establish the connection between API and Database using Express – My SQL driver
- **Introduction to My SQL**
- Write a program to create a Database and table inside that database using My SQL Command line client
- Write a My SQL queries to create table, and insert the data, update the data in the table
- Write a My SQL queries to implement the subqueries in the My SQL command line client
- Write a My SQL program to create the script files in the My SQL workbench
- Write a My SQL program to create a database directory in Project and initialize a database. sql file to integrate the database into API

Textbooks:

- Web Design with HTML, CSS, JavaScript and JQuery Set Book by Jon Duckett Professional JavaScript for Web Developers Book by Nicholas C. Zakas
- John Dean, Web Programming with HTML5, CSS and JavaScript, Jones & Bartlett Learning, 2019.
- Pro MERN Stack: Full Stack Web App Development with Mongo, Express, React, and Node, Vasan Subramanian, 2nd edition, APress, O’Reilly.
- Learning PHP, MySQL, JavaScript, CSS & HTML5: A Step-by-Step Guide to Creating Dynamic Websites by Robin Nixon
- AZAT MARDAN, Full Stack Java Script: Learn Back bone. js, Node.jsand Mongo DB.2015

Reference Books:

- Full-Stack JavaScript Development by Eric Bush.
- Programming the World Wide Web, 7th Edition, Robet W Sebesta, Pearson, 2013.
- Tomasz Dyl , Kamil Przeorski , Maciej Czarnecki, Mastering Full Stack React Web Development 2017

Online Learning Resources:

- https://ict.iitk.ac.in/product/full-stack-developer-html5-css3-js-bootstrap-php-4/
- https://www.w3schools.com/html
- https://www.w3schools.com/css
- https://www.w3schools.com/js/
- https://www.w3schools.com/nodejs
- https://www.w3schools.com/typescript